Turbulence Analysis Report - 2026-02-22

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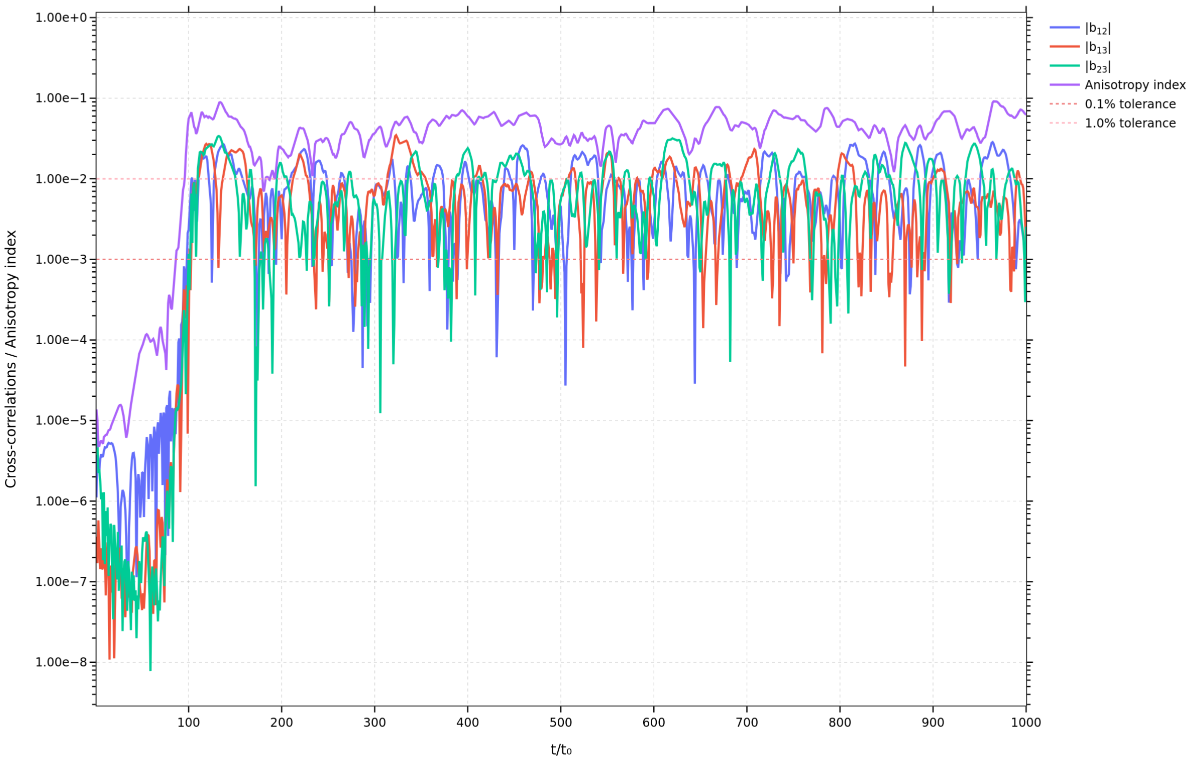

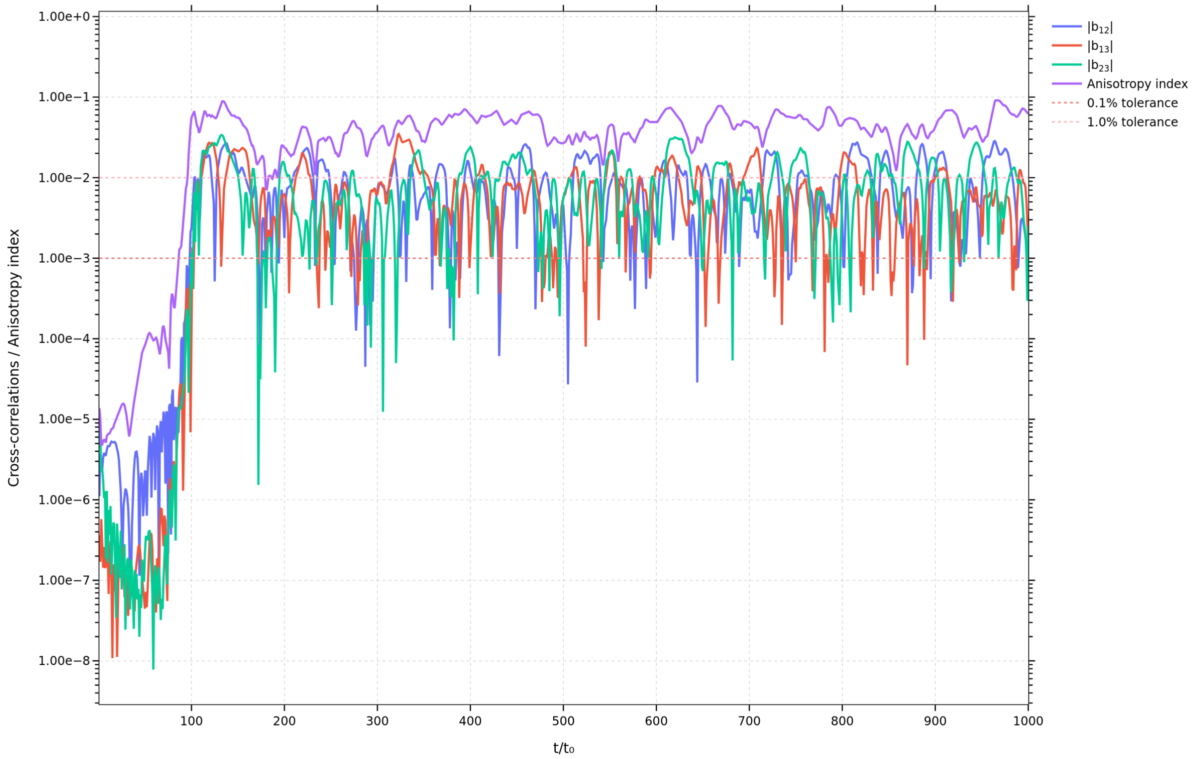
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Anisotropy index
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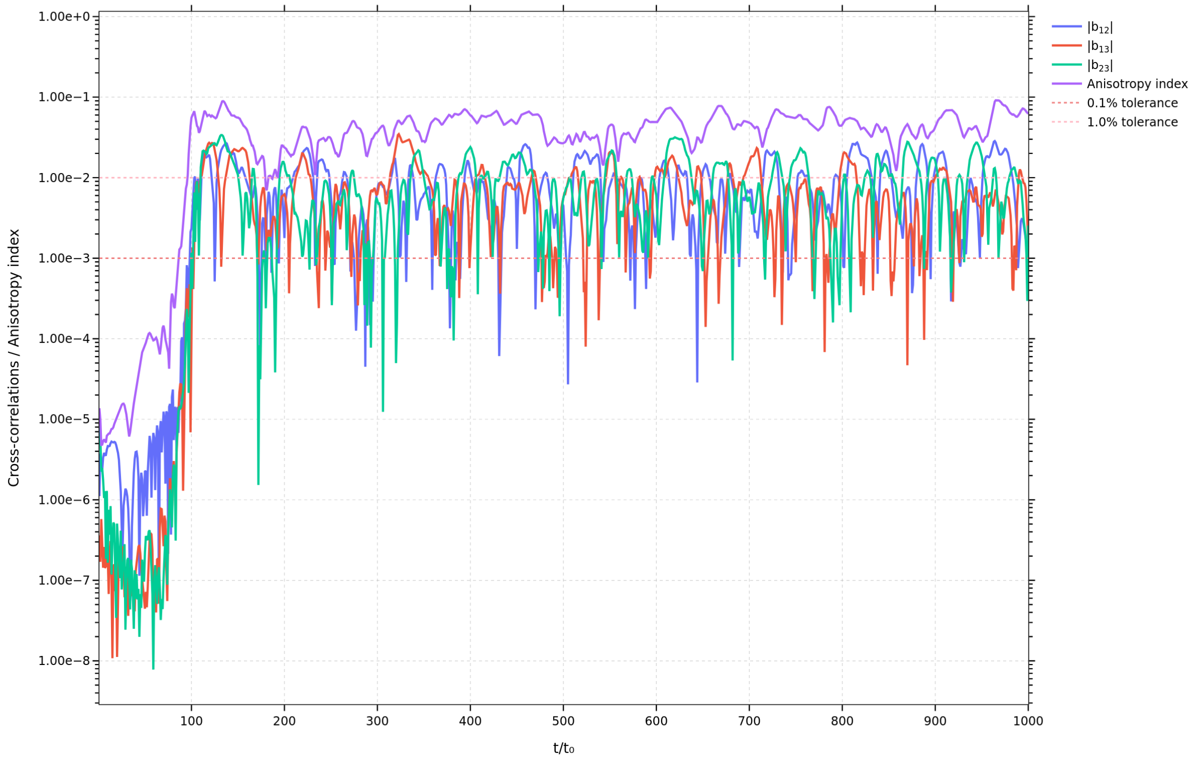
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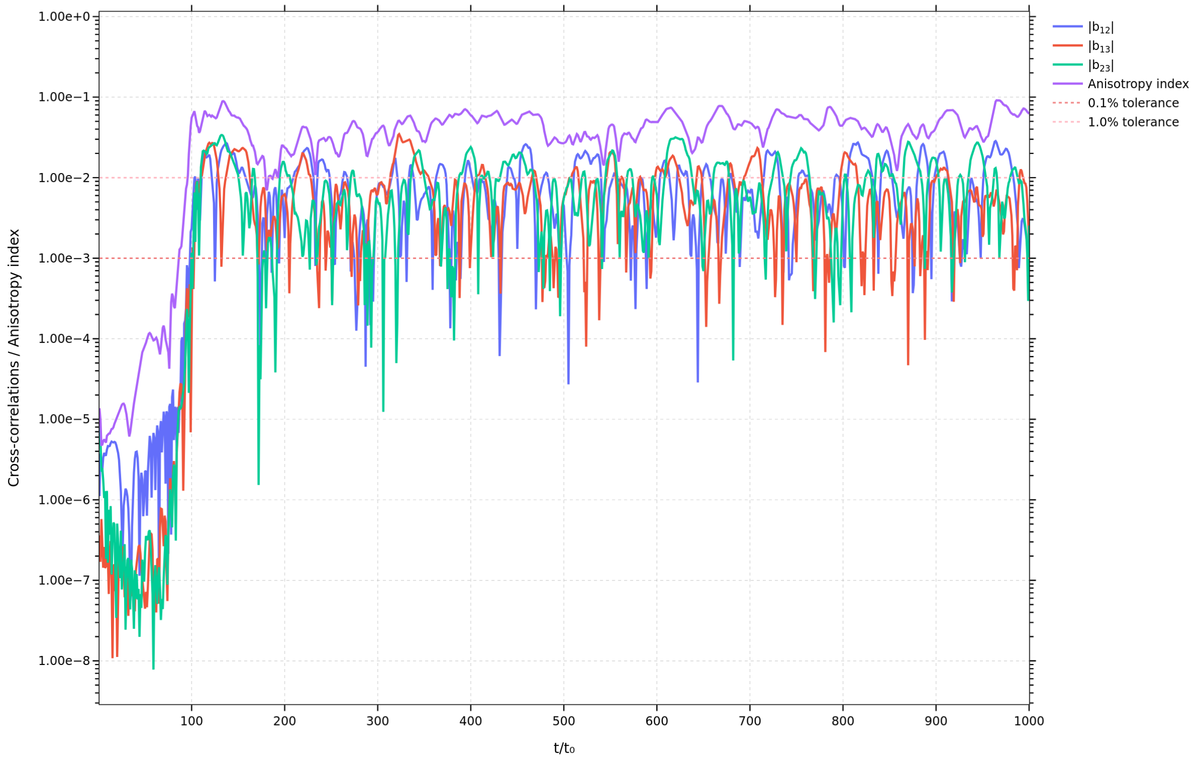
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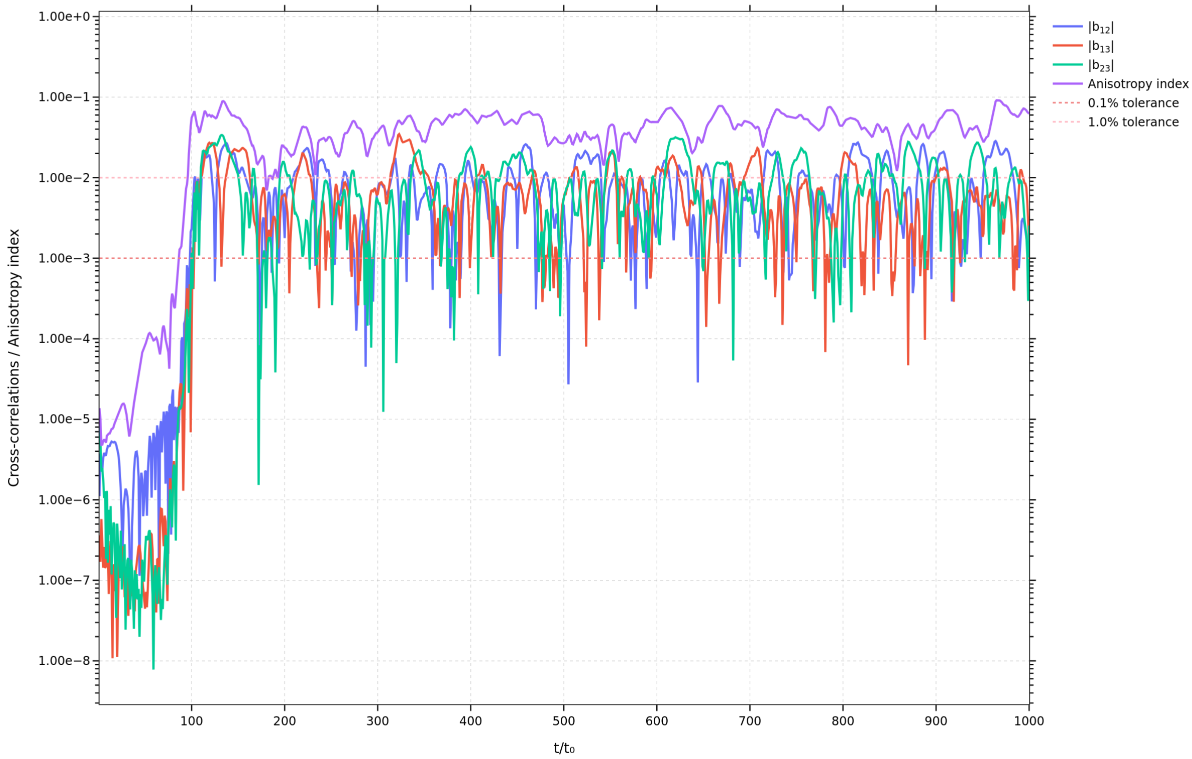
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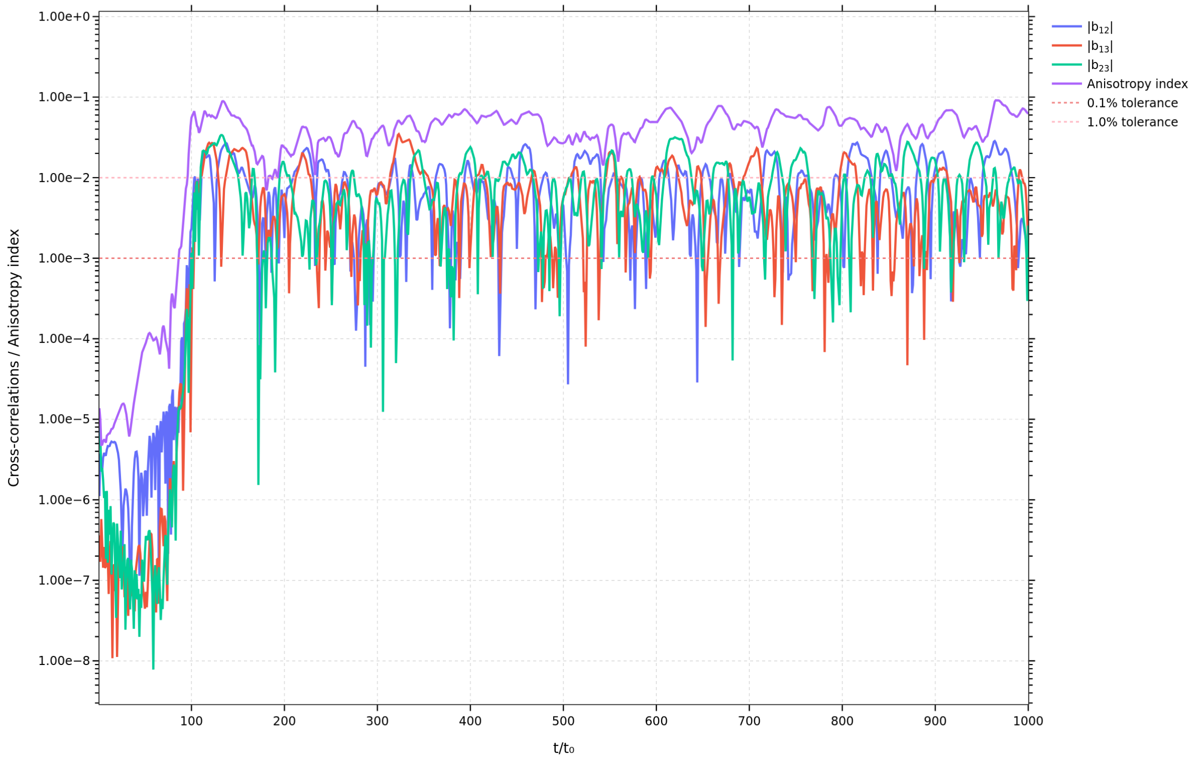

|b12|
|b13|
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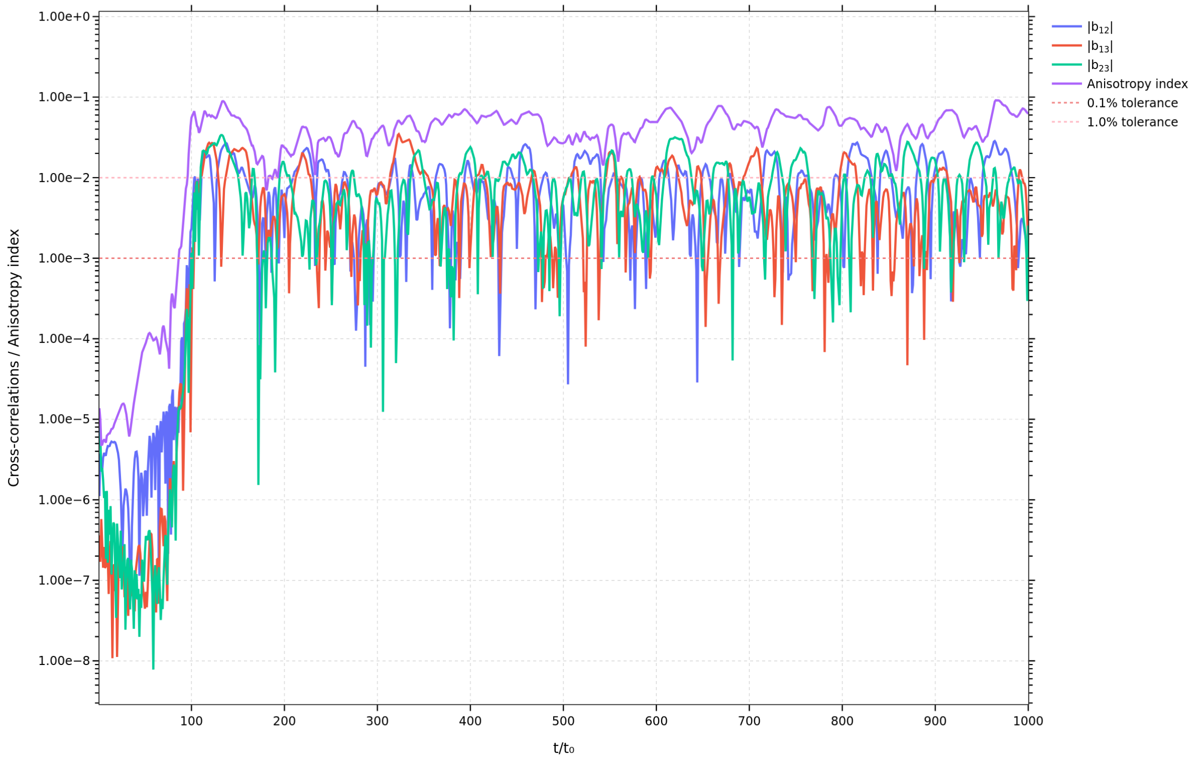
Anisotropy index
0.1% tolerance
1.0% tolerance
Cross-correlations / Anisotropy index
t/t0



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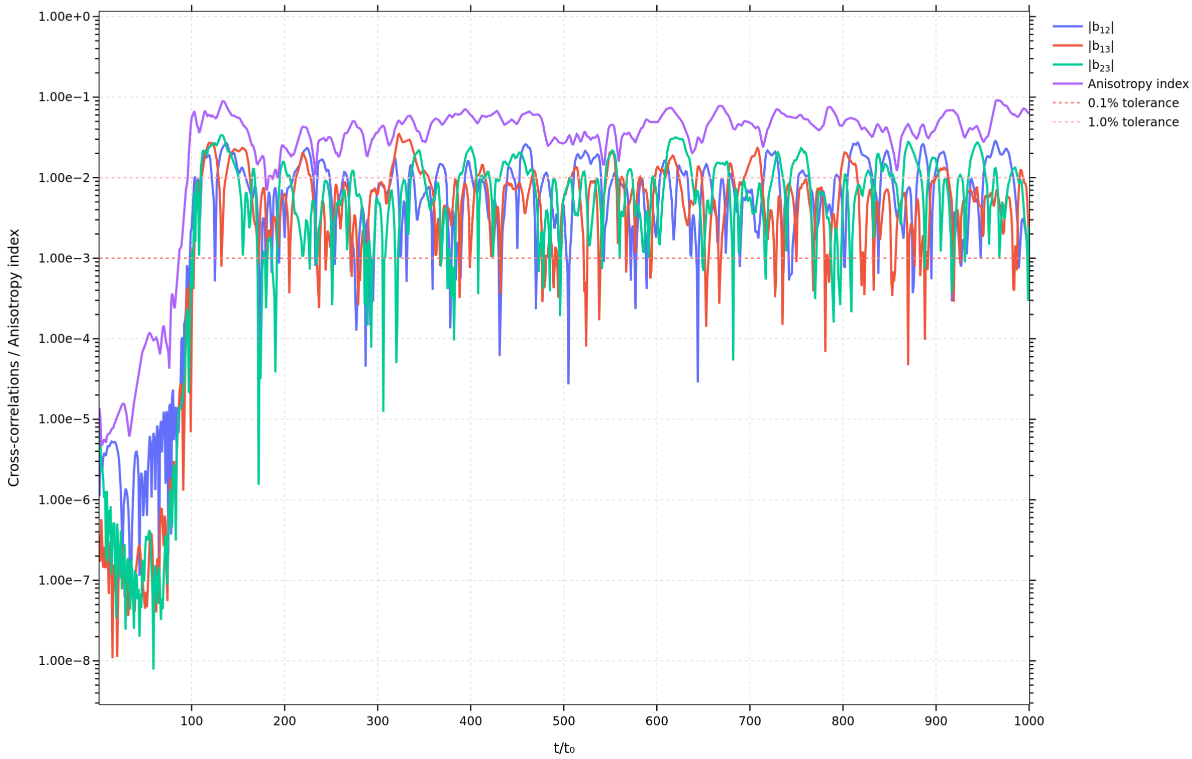
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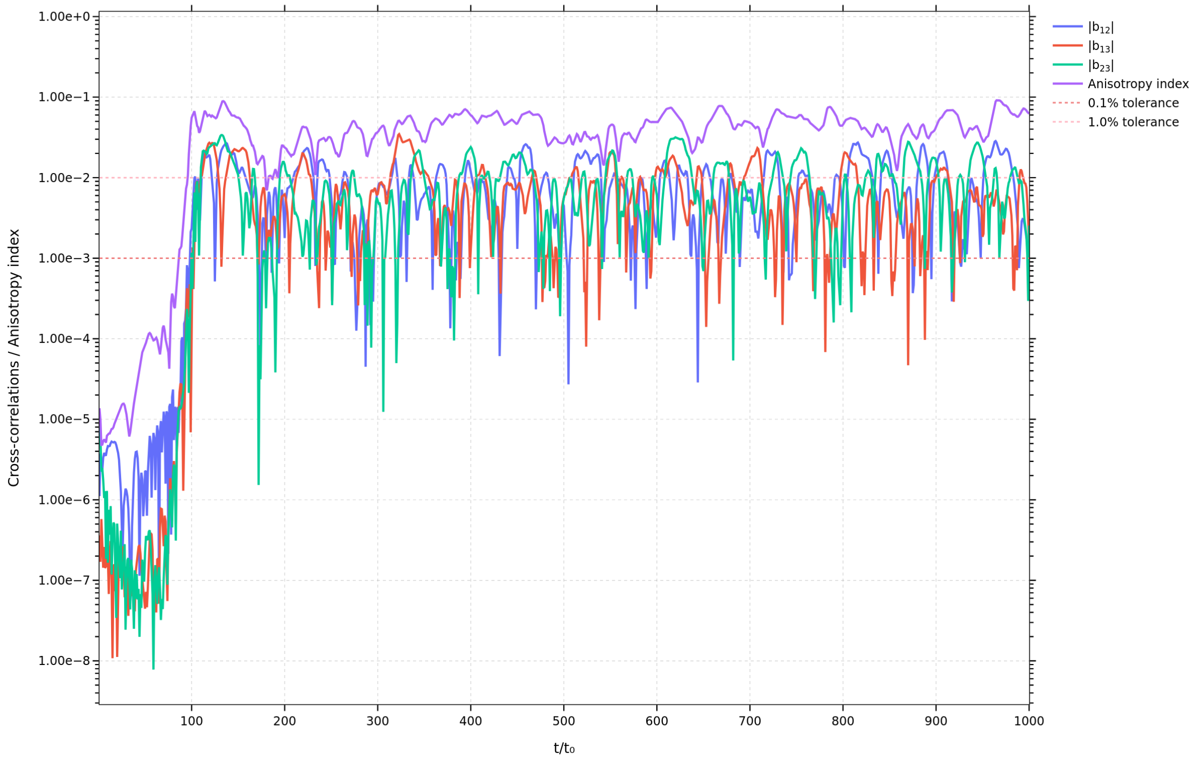
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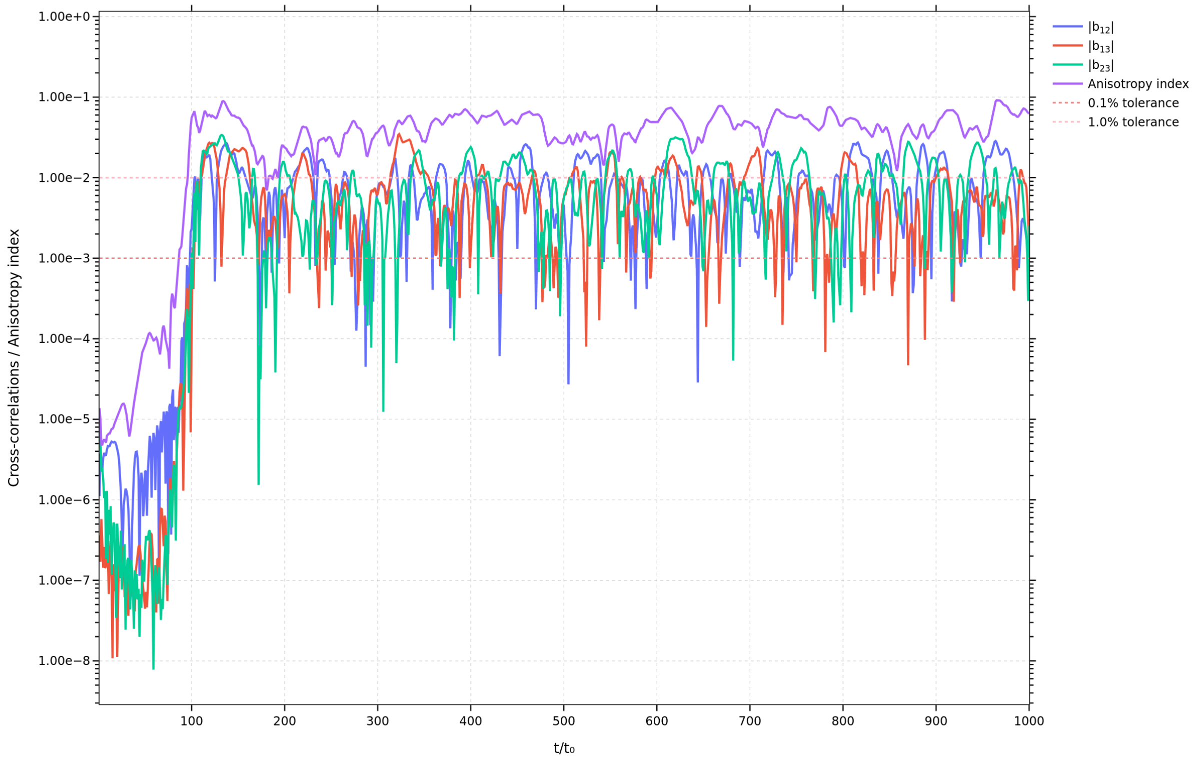
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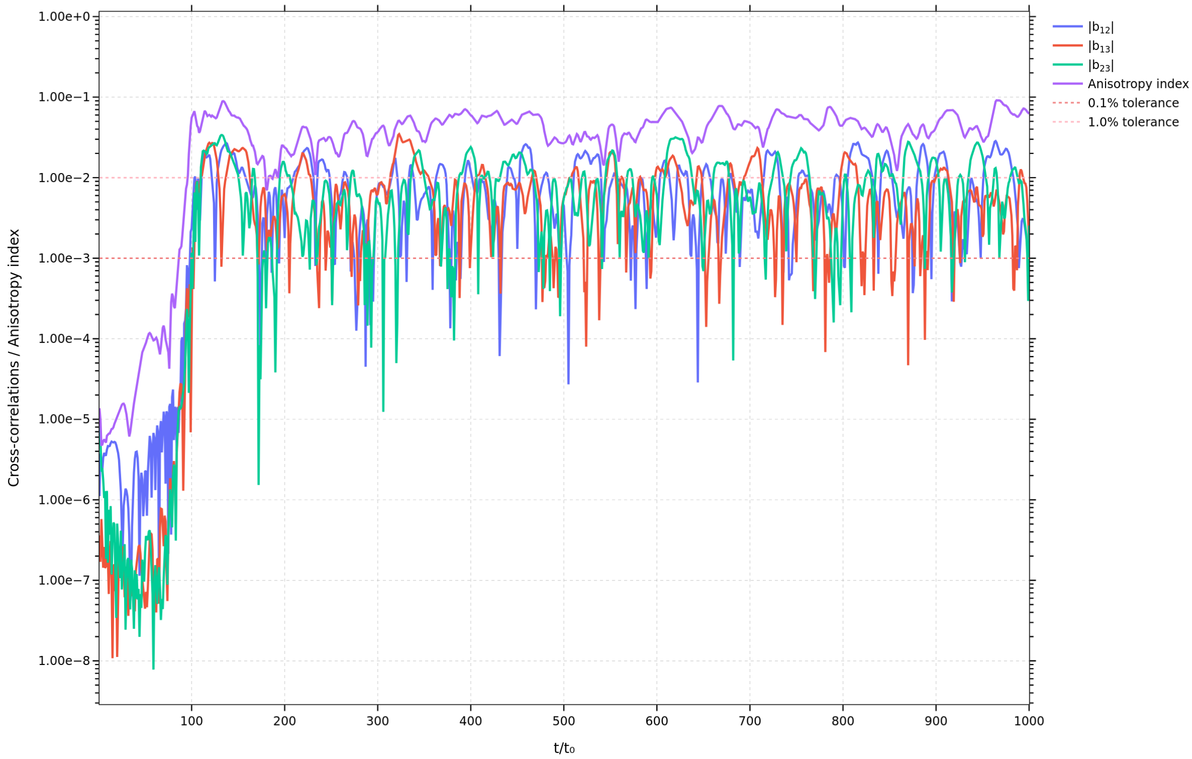
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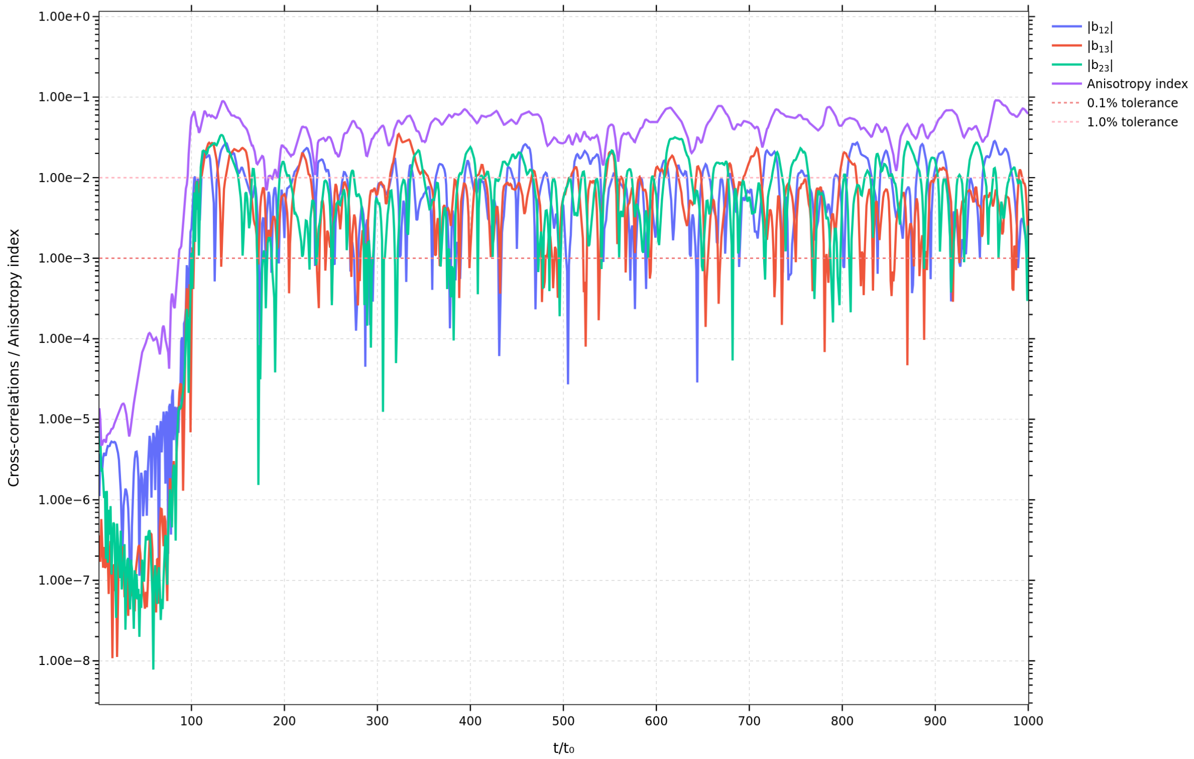

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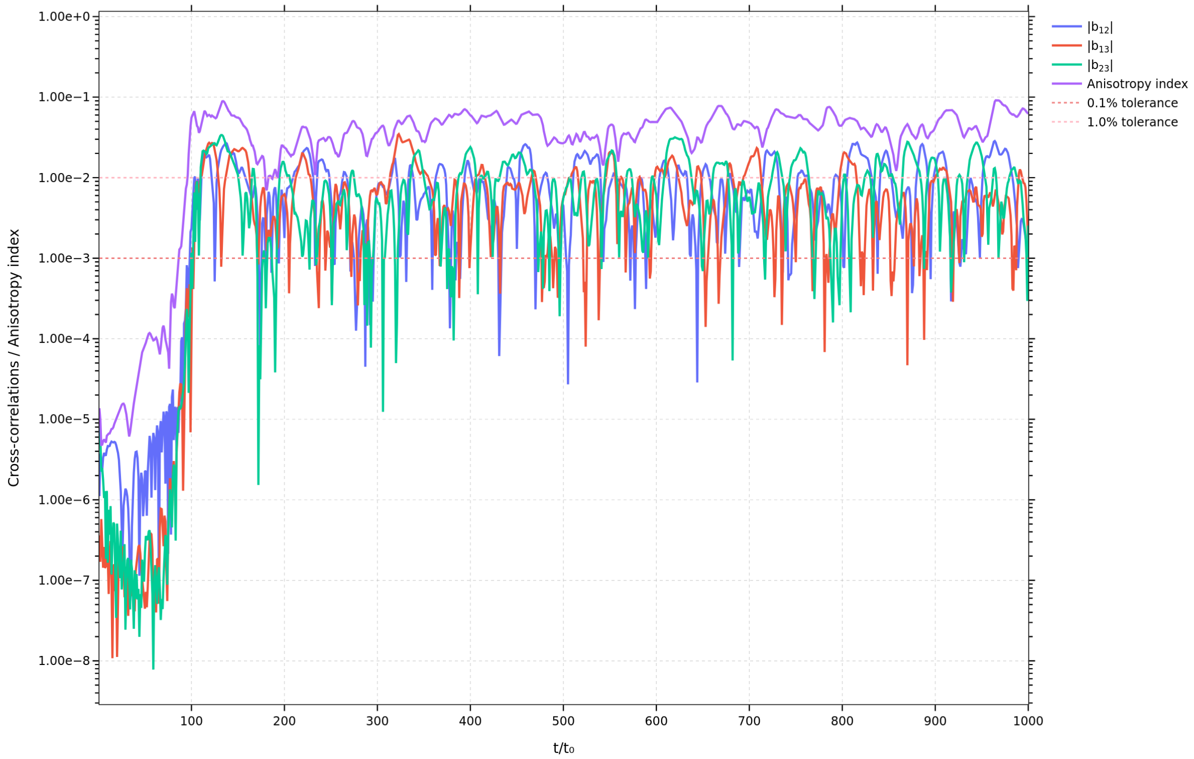

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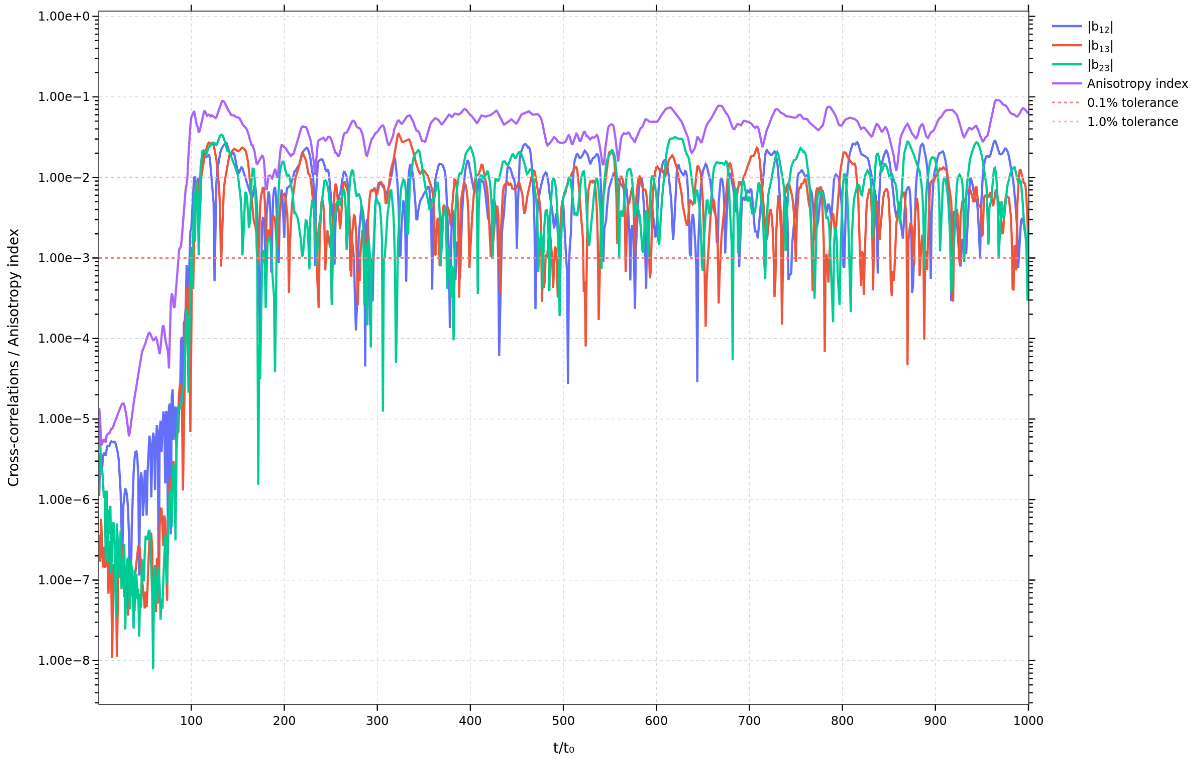

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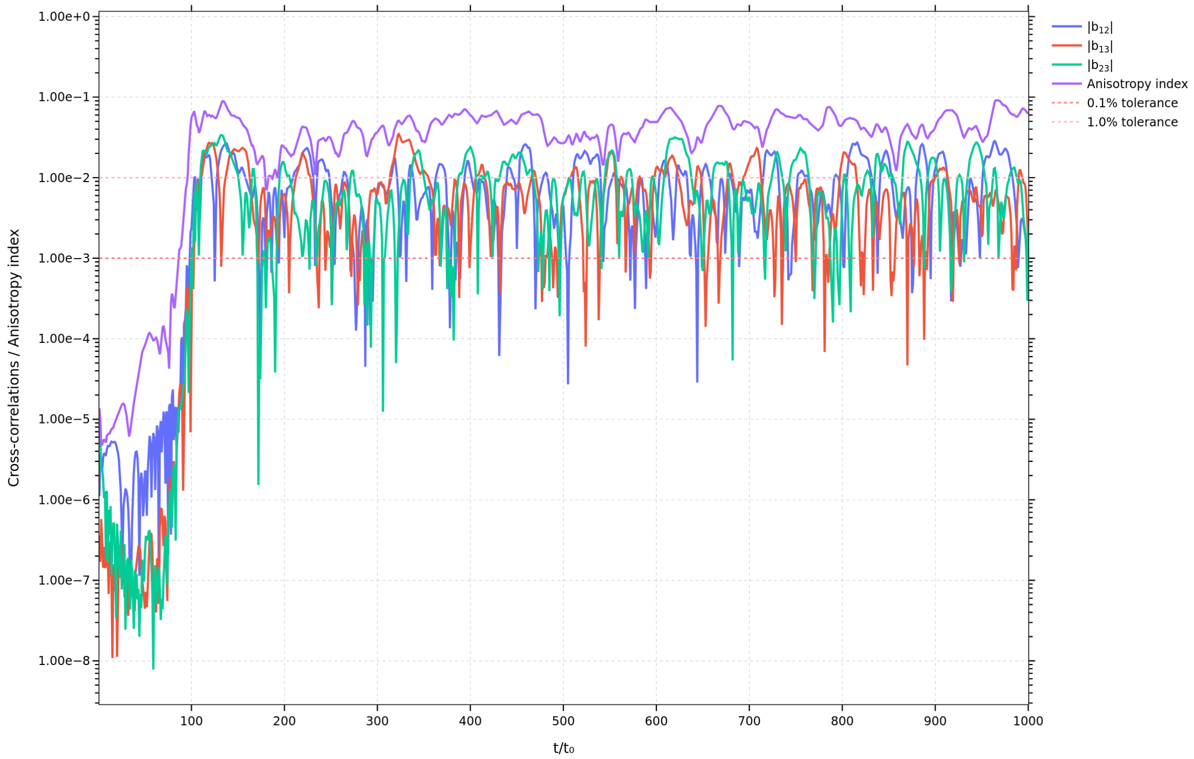

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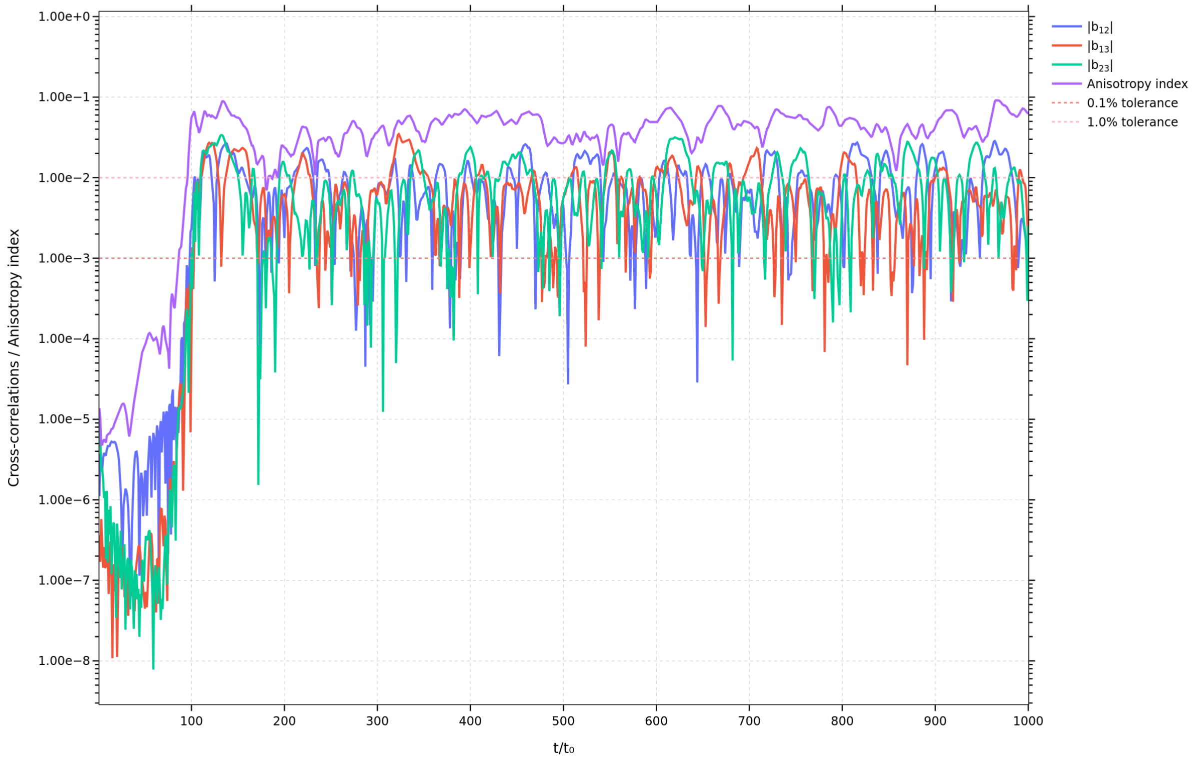

Figure 16. This plot shows the cross-correlations |b12|, |b13|, |b23|, and the anisotropy index versus normalized time t/t0.

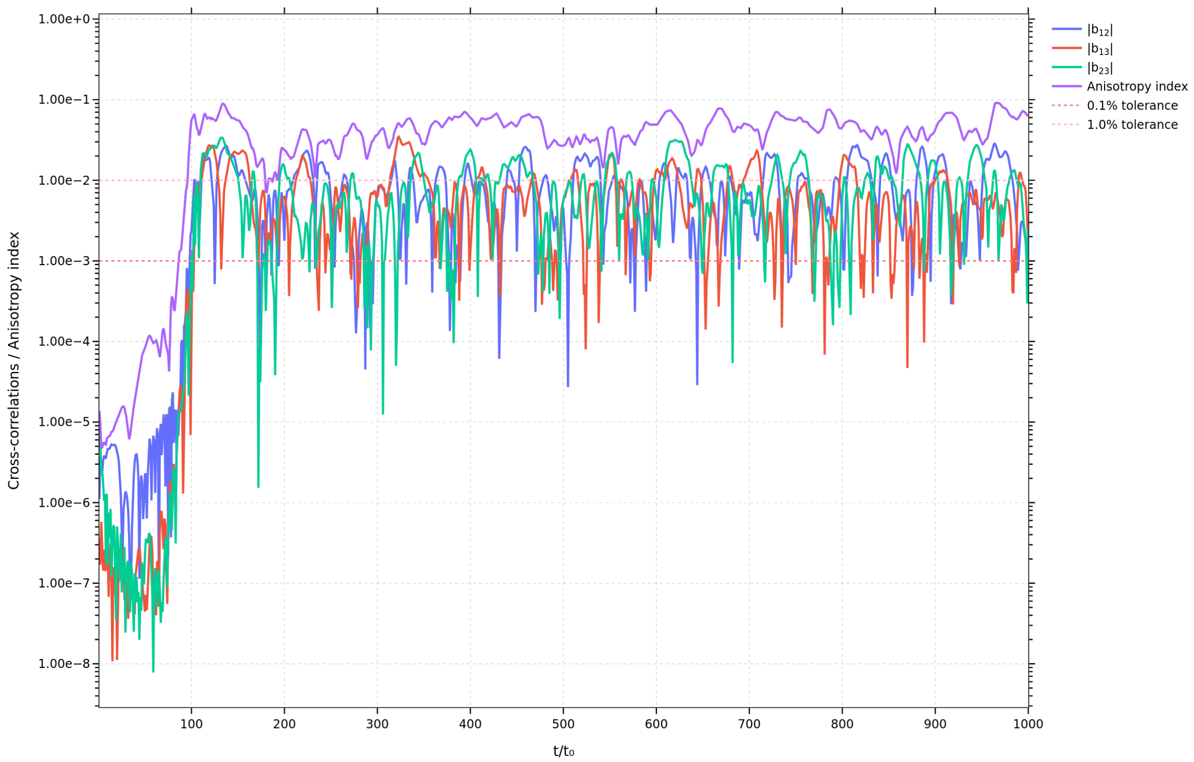

Figure 17. This plot shows the cross-correlations |b12|, |b13|, |b23|, and the anisotropy index versus normalized time t/t0.

Explanation of Cross-correlations Figure

This figure, titled "Cross-correlations / Anisotropy Index vs. t/t_0", displays the temporal evolution of key metrics used to quantify turbulence anisotropy. The x-axis represents normalized time (t/t_0), while the y-axis, on a logarithmic scale, shows the magnitudes of the off-diagonal components of the anisotropy tensor and a scalar anisotropy index.

Metrics Represented:

1. $|b_{12}|$**,** $|b_{13}|$**,** $|b_{23}|$: These traces (blue, red, and green lines, respectively) represent the magnitudes of the off-diagonal components of the Reynolds stress anisotropy tensor, b_{ij}. The anisotropy tensor quantifies the deviation of the Reynolds stress tensor from an isotropic state. For perfectly isotropic turbulence, all off-diagonal components of b_{ij} are zero. Therefore, non-zero values of $|b_{ij}|$ indicate the presence and magnitude of anisotropy in the respective planes (e.g., $|b_{12}|$ relates to anisotropy in the 1-2 plane).
2. **Anisotropy index**: This trace (purple line) is a scalar measure that provides an overall quantification of the flow's anisotropy. It typically combines information from all components of the anisotropy tensor into a single value. A value of zero for the anisotropy index signifies perfect isotropy. Higher values indicate a greater degree of anisotropy.
3. **0.1% tolerance (dashed red line) and 1.0% tolerance (dashed pink line)**: These horizontal lines serve as thresholds to assess the approach to isotropy. If the anisotropy index or the cross-correlation magnitudes fall below these lines, it suggests that the turbulence is approaching a state of near-isotropy within the specified tolerance.

Interpretation of Trends:

- **Initial Phase (t/t_0 < ~100)**: In the very early stages of the simulation (up to approximately $t/t_0 = 100$), all metrics ($|b_{12}|$, $|b_{13}|$, $|b_{23}|$, and the Anisotropy index) are very low, indicating an initial state that might be close to isotropic or a developing flow.

- **Development of Anisotropy (t/t_0 from ~100 to ~200)**: Following the initial phase, there is a sharp increase in all anisotropy metrics. The Anisotropy index (purple line) rises significantly, reaching values around 10^{-1} to 10^{-2}. The individual cross-correlations also increase, fluctuating between 10^{-3} and 10^{-2}. This indicates that the turbulence quickly develops significant anisotropy.
- **Fluctuating Anisotropy (t/t_0 > ~200)**: For the remainder of the simulation, the Anisotropy index generally fluctuates between approximately 2×10^{-2} and 10^{-1}. It consistently remains above both the 0.1% (10^{-3}) and 1.0% (10^{-2}) tolerance lines, suggesting that the flow, on average, maintains a notable level of anisotropy and does not achieve a state of near-isotropy within these tolerances over sustained periods.
- **Cross-correlation Fluctuations**: The individual cross-correlation magnitudes ($|b_{12}|$, $|b_{13}|$, $|b_{23}|$) show significant, rapid fluctuations. While they often dip below the 1.0% tolerance line and occasionally even below the 0.1% tolerance line, they also frequently peak above these thresholds. This oscillatory behavior indicates that while certain components of anisotropy might temporarily decrease, the overall anisotropic character of the flow persists. The fact that the Anisotropy index (an aggregate measure) remains higher than the individual cross-correlations for most of the simulation reinforces this observation.

In summary, the figure demonstrates that after an initial development phase, the turbulent flow exhibits persistent anisotropy, with the overall anisotropy index remaining above the specified tolerance levels throughout the observed simulation time. While individual cross-correlation components show transient reductions, the flow does not consistently approach a state of isotropy.